

1 Solution — GIAM 1.3

Problem 4

1.1 Problem

Write your own sentences containing four quantifiers. One sentence in which the quantifiers appear $(\forall\exists\forall\exists)$ and another in which they appear $(\exists\forall\exists\forall)$.

1.2 Sentence A — Pattern $\forall\exists\forall\exists$

“For every city, there is an airport from which every traveler can catch some flight.”

Reading left to right, the four quantifier phrases are:

Phrase	Quantifier	Type
“every city”	$\forall$	universal
“an airport”	$\exists$	existential
“every traveler”	$\forall$	universal
“some flight”	$\exists$	existential

Table 1.1

So the pattern is $\forall\exists\forall\exists$. Symbolically, with **Serves**(a, c) for “airport a serves city c ”, **From**(f, a) for “flight f departs from a ”, and **Catch**(t, f) for “traveler t can catch flight f ”:

$$\forall c \exists a \left[\text{Serves}(a, c) \wedge \forall t \exists f (\text{From}(f, a) \wedge \text{Catch}(t, f)) \right].$$

The quantifier prefix reads $\forall c \exists a \forall t \exists f$ — exactly $\forall\exists\forall\exists$.

1.3 Sentence B — Pattern $\exists\forall\exists\forall$

“There is a website on which every student can find a tutorial that every beginner can follow.”

The four quantifier phrases, in order:

Phrase	Quantifier	Type
“a website”	$\exists$	existential
“every student”	$\forall$	universal
“a tutorial”	$\exists$	existential
“every beginner”	$\forall$	universal

Table 1.2

So the pattern is $\exists\forall\exists\forall$. Symbolically, with $\text{On}(u, w)$ for “tutorial u is on website w ”, $\text{Find}(s, u)$ for “student s can find u ”, and $\text{Follow}(b, u)$ for “beginner b can follow u ”:

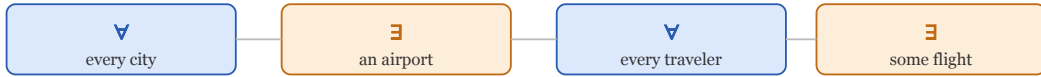
$$\exists w \forall s \exists u \left[\text{On}(u, w) \wedge \text{Find}(s, u) \wedge \forall b \text{Follow}(b, u) \right].$$

The prefix reads $\exists w \forall s \exists u \forall b$ — exactly $\exists\forall\exists\forall$.

1.4 The Two Patterns Side by Side

Sentence A — pattern $\forall \exists \forall \exists$

"For every city, there is an airport from which every traveler can catch some flight."



Sentence B — pattern $\exists \forall \exists \forall$

"There is a website on which every student can find a tutorial that every beginner can follow."



□ $\forall$ universal □ $\exists$ existential

Figure 1.1: Sentence A's quantifier phrases tag as $\forall \exists \forall \exists$ (every city, an airport, every traveler, some flight); Sentence B's tag as $\exists \forall \exists \forall$ (a website, every student, a tutorial, every beginner). Universal quantifiers are blue, existential orange.

1.5 Remark — A Recipe for Building Such Sentences

There is a reliable template. Pick four nouns and chain them, alternating the determiner that signals each quantifier:

- $\forall$ is signalled by "every / for every / all / each";
- $\exists$ is signalled by "a / an / some / there is".

Then nest them so each quantifier sits inside the scope of the previous one. For $\forall \exists \forall \exists$:

$$\underbrace{\text{For every } N_1}_{\forall}, \underbrace{\text{there is a } N_2}_{\exists} \text{ such that } \underbrace{\text{every } N_3}_{\forall} \underbrace{\text{has some } N_4 \dots}_{\exists}$$

Swapping the opening determiner from “every” to “there is a” flips the whole pattern's parity — which is exactly how Sentence B ($\exists \forall \exists \forall$) is built from the same skeleton as Sentence A.

1.6 Remark — Order Is Not Cosmetic: $\forall \exists$ vs $\exists \forall$

The point of the exercise — and of the whole section — is that the *order* of quantifiers controls meaning. Consider just the first two quantifiers of each sentence:

- Sentence A opens $\forall c \exists a$: “for every city there is an airport” — each city gets its *own* (possibly different) airport.
- Sentence B opens $\exists w \forall s$: “there is a website such that every student...” — *one single* website serves all students.

These are genuinely different kinds of claim. The textbook’s own cafeteria example makes the same contrast:

- “ $\forall$ student $\exists$ food they like” — everyone has *a* favourite (possibly different).
- “ $\exists$ food $\forall$ student likes it” — *one* food pleases everyone (much stronger).

In a four-quantifier sentence this sensitivity compounds at *each* adjacent $\forall\exists$ / $\exists\forall$ junction — which is why being able to *write* a sentence to a prescribed quantifier pattern, as this exercise asks, is a genuine skill and not just notation-shuffling.

1.7 Remark — A Purely Mathematical Pair

Natural-language sentences are fine here, but the same patterns occur in clean mathematical statements — where they can be checked for truth. Over the integers $\mathbb{Z}$:

$\forall\exists\forall\exists$ (true):

$$\forall m \exists n \forall k \exists j \ (m + j = n + k).$$

Why true: given m , choose $n = m$; then for each k , choose $j = k$, so $m + j = m + k = n + k$. ✓

$\exists\forall\exists\forall$ (true):

$$\exists z \forall x \exists y \forall t \ ((x + y) \cdot t = z).$$

Why true: choose $z = 0$; for each x , choose $y = -x$, so $x + y = 0$ and $(x + y) \cdot t = 0 = z$ for *every* t . ✓

These show the patterns in their natural mathematical habitat: each is a nest of “for all / there exists” where the existential witnesses (n, j in the first; y in the second) are chosen *as functions of* the universally-quantified variables that precede them.

1.8 Remark — Nesting and Scope

In both sentences the quantifiers are **nested**: each one lies inside the scope of all the quantifiers to its left. The practical consequence is that an inner existential witness may depend on *every* outer variable. In Sentence A, the flight f (innermost $\exists$) can depend on *both* the city c and the traveler t ; the airport a (second $\exists$) can depend on the city c . Reading the prefix $\forall c \exists a \forall t \exists f$ left to right is the same as reading nested dependencies: *choose c adversarially, respond with a , then c 's opponent chooses t , and we respond with f* . This “game” reading — universal = adversary’s move, existential = our response — is the cleanest way to keep a four-quantifier sentence’s meaning straight, and it is why the *order* of the moves is everything.

1.9 Remark — There Are Many Right Answers

This is an open-ended exercise: *any* grammatical sentence whose four quantifier words appear in the prescribed order is a correct answer. The two given above are only samples. A reader’s own pair — “For every lock there is a key that every locksmith can copy for some customer” ($\forall \exists \forall \exists$), say, or “There is a law that every citizen has a right which every court must protect” ($\exists \forall \exists \forall$) — is equally valid. What is being assessed is solely that the quantifier *types* occur in the requested *sequence*; the subject matter is free.